5.1 1, 3, 33

Hw. 13, 20

## § 5.2 Characteristic equations

Q How to find the eigenvalues?

Exmp. 1 Find the eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$.

Sol We need to find all scalars $\lambda$ s.t. the matrix equ.

$$(A - \lambda I)\vec{x} = 0 \qquad (*)$$

has a nontrivial solution. By the Invertible Matrix Theorem in § 2.3, the equation has a nontrivial sol. iff $A - \lambda I$ is NOT invertible, iff $A - \lambda I$ has determinant 0.

So eigenvalues of $A$ are the sol. of the equation

$$\boxed{\det(A - \lambda I) = 0}$$

On the other hand

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{vmatrix} = (2-\lambda)(-6-\lambda) - 9$$

$$= -12 + 4\lambda + \lambda^2 - 9 = \lambda^2 + 4\lambda - 21$$

$$= (\lambda + 7)(\lambda - 3)$$

Thus, eigenvalues of $A$ are $-7$ or $3$. ☐

Summarize. Characteristic

The eigenvalues of the matrix $A$ are the solutions of the equation

$$\det(A - \lambda I) = 0 \qquad (\lambda \rightsquigarrow \text{the unknown})$$

Thm (The Invertible Matrix Theorem) (cont'd)

$A$: $n \times n$ matrix. Then $A \rightsquigarrow$ inv. iff

(r) $0 \rightsquigarrow$ NOT an eigenvalues of $A$.

Explain $A$ inv. $\Leftrightarrow A\vec{x} = \vec{0}$ has only trivial sol.

$= 0 \cdot \vec{x}$.

Exmp 3 Find the eigenvalues of $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Sol Characteristic equation:

$$\det(A - \lambda I) = \det\begin{bmatrix} 5-\lambda & -2 & 6 & -1 \\ & 3-\lambda & -8 & 0 \\ & & 5-\lambda & 4 \\ & & & 1-\lambda \end{bmatrix}$$

$$= (5-\lambda)(3-\lambda)(5-\lambda)(1-\lambda) = 0$$

Eigenvalues of $A$: 5, 3, 1. (1a)

(Generalization)

Thm 1 The eigenvalues of a triangular matrix are the entries on its main diagonal.

Def $A$: $n \times n$ matrix

The characteristic polynomial of $A$ is

$$\det(A - \lambda I)$$

which is a polynomial in $\lambda$ of degree $n$.

The (algebraic) multiplicity of an eigenvalue $\lambda$ is its multiplicity as a root of the char. poly / equation.

Exmp In exmp 3, the multiplicity of 5 is 2.

## Similarity

It is easy to get the eigenvalues of triangular matrices, in particular diagonal matrices.

Q Can we relate a matrix to diagonal matrix?

**Def** $A$ & $B$ : $n \times n$ matrices

If $\exists$ invertible $P$ s.t. $P^{-1}AP = B$ (or equivalently $A = PBP^{-1}$)

then $A$ & $B$ are **similar**.

Changing $A$ into $P^{-1}AP$ is called a **Similarity transformation**.

**Thm 4** If $A$ & $B$ are similar, then they have the same char. poly and hence the same eigenvalues. (w/ the same multiplicity)

**Pf** Suppose $P^{-1}AP = B$

char poly of $B$: $\det(B - \lambda I) = \det(P^{-1}AP - \lambda I)$

$= \det(P^{-1}AP - P^{-1}(\lambda I)P) =$

$= \det(P^{-1}(A - \lambda I)P)$

$= \det(P^{-1}) \det(A - \lambda I) \det(P)$

$= \det(P^{-1}) \cdot \det(P) \det(A - \lambda I)$

$= \det(A - \lambda I)$ char poly of $A$ ☐

That is, similarity transformations preserve eigenvalues.

**Exmp** If $P^{-1}AP = B$ then $B^k = P^{-1}A^kP$ ☐

↑ This is easy if $A$ is a diagonal matrix.

These leads to "diagonalization".